

Pattern Matching and Machine Learning for Audio Signal Processing

Exercise sheet 5

To be uploaded in eCampus till: 05-06-2026 22:00 (strict deadline)

Exercise 5.1

[3 + 3 + 4 + 2 = 12 points]

In this exercise we consider the cyclic convolution and the cyclic autocorrelation as defined in the lecture in Chapter 2.

- (a) Calculate by hand the cyclic autocorrelation of the vector $x = [4, 1, 2, 0, 1]$.
- (b) Calculate by hand the cyclic convolution of the vectors $x = [4, 1, 2, 0, 1]$ and $y = [1, 3, 2, 0, 1]$.
- (c) Create a function in Matlab / Python that calculates and plots the cyclic convolution of two signals and test it with the signals x and y given in (b).
- (d) Extend your program such that the linear convolution of two discrete signals (defined in slide 15 of chapter 3), both having finite support, can be computed using cyclic convolution. In this you may assume that both signals are given as vectors x and y in Matlab / Python. Test your function with the vectors $x = [1, 3, 2]$ and $y = [2, 3]$.

Exercise 5.2

[3 + 1 + 1 + 5 = 10 points]

This exercise introduces the idea of the short-time Fourier Transform (STFT), as you have seen it in the lecture.

- (a) Create a function that generates a concatenation of several sine waves with different frequencies and amplitudes. The frequencies, the amplitudes, and the sampling frequency of the sine waves have to be provided as input to the function.
- (b) Use the function you created in (a) to generate a signal that consists of a concatenation of two sine waves of 3 seconds length each and with frequency 1 Hz and 5 Hz respectively. The sampling frequency should be 5 kHz and the resulting signal should have a length of 6 seconds.
- (c) Create a Hann window of 2 seconds length. The sampling frequency should be the same as in part (b).

- (d) The idea of the STFT is that you do not analyze the whole signal at once with the Fourier transform, but window by window. In this part of the exercise you have to create a script which implements this process. Move the window function you created in (c) along the signal (created in (b)) without overlapping the windows and perform a pointwise multiplication of the section of the signal and the window function. Then calculate the Fourier transform of the resulting product. For each step plot in the same figure the signal, the window and the resulting product. Furthermore visualize the magnitude spectrum for each step. Make use of subplots. Your plots should look similar to the ones of Chapter 2 slide 34.

Exercise 5.3

[6 points]

Create a Matlab / Python function which implements the denoising procedure presented in the lecture and use it on the signal defined in Exercise 4.2.